Marmara University Electrical-Electronics Eng. Dept.

CSE2003 Digital Design Laboratory Assignment #5

The studies will be uploaded to the system before due date (next wedesday). **Each group must present their own study. The group members have to work together and each member must know each step of the study.**

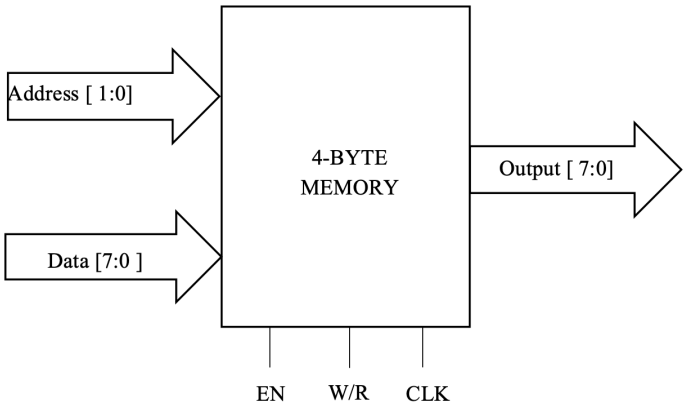
For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design a hardware floating point multiplier that has the Input/output defiitions below.

The steps you need to follow: 1. You should design the circuit, **2.**write the verilog HDL code fort he designed circuit, **3.** Check if the designed hardware is correctly coded by using RTL viewer. Don't forget: your circuit may look different then yours since the compiler can make some optimizations. Please take this information into account when you compare the RTL viewer circuit and yours. **4.** Verify your design by using simulation tool (you need to check the inputs and outputs). **5.** implement and RUN your code using Altera DE-115 FPGA kit. Use the remote desktop connection (**Remote desktop connection ID**:256 521 711 **password**: dijitallab2021).

You need to upload the following files before due date:

1. A short video (<60 secs.) that shows all steps of the study. Especially, the implementation part must be included.
2. The report (in PDF format): 1. What is the purpose of the study? 2. What are the design details with necessary explenations. 3. The design and implementation steps (all steps with screenshots etc.). 4. The result and coclusion: did you reach the purposed result? If the answer is no, why? if it is yes, how?

For this laboratory assignment, you will use Altera Quartus II software and DE-115 FPGA kit and design an 4-byte RAM (Memory) that has the Input/output definitions below:



- EN signal enables the module, The output must be in high impedance state if EN=0.

- W/R signal defines the operation mode. Output shows the data in the given adrress when W/R =1 (Read mode). The Data will be stored to the Address location if W/R = 0.

- CLK is the pulse signal which should be connected to Flip Flops. You may use the pushbutton as a controllable clock signal to test your device is working properly.

- 4 Byte will be addressed using 2-bit Address input.

- Each Adress stores 8-bit data.

- You are expected to test your device and to demonstrate it by using the output LEDs.

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Hint: Each address should be built using Flip-Flops. Each Flip-flop stores 1-bit data. Use a DeMUX to select the required address.

1. Design a Flip Flop module (i.e: D-FF.), …USE GATE LEVEL MODELING (e.g: c=a&b ✗, and(c,a,b)✓)…
2. Make the Flip-Flops connections in order to built 8-bit data memory.
3. Design a DEMUX module, …USE GATE LEVEL MODELING (e.g: c=a&b ✗, and(c,a,b)✓)…
4. Design the 4-byte memory module as a top-level module (This module will use the Flip-Flop modules and DEMUX modules)
5. Use the 4-byte memory module and derive the FPGA code.
6. Synthesize the circuit, program the FPGA, and verify its operation.